

GULF SHORES, AL, USA

WMO: 720381

Lat: 30.291N

Lon: 87.672W

Elev: 16

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63885

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	32.9	12.2	10.2	37.8	18.4	13.9	41.1	22.3	59.5	20.0	60.7	3.7	0	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	93.5	78.6	91.3	77.9	90.2	77.7	82.4	89.0	81.3	87.5	80.3	86.3	8.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.2	85.9	79.6	153.5	84.6	78.7	149.2	83.9	46.1	89.7	44.8	87.3	43.8	86.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.9	15.3	24.6	96.1	2.7	3.8	22.6	98.8	21.0	101.0	19.4	103.1	17.5	105.9
			22.6	82.9	2.7	2.0	20.6	84.4	19.1	85.5	17.6	86.7	15.6	88.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	52.1	55.0	60.8	66.4	73.7	80.6	81.9	78.7	69.5	59.2	56.0
		DBStd	12.67	9.52	8.94	7.73	5.91	5.03	3.62	2.93	3.10	3.88	7.41	8.00
		HDD50	211	89	49	11	1	0	0	0	0	0	15	46
		HDD65	1485	409	294	172	53	6	0	0	0	42	207	302
		CDD50	6796	155	189	345	491	735	919	987	988	860	605	291
		CDD65	2595	10	14	41	94	275	469	522	523	410	182	34
		CDH74	25055	2	17	87	423	2281	5220	5826	5738	3789	1519	135
		CDH80	9292	0	0	2	15	525	2184	2460	2403	1345	353	5
(14)	Wind	WSAvg	6.1	6.7	7.0	7.6	7.3	6.6	5.8	4.9	4.8	5.0	5.9	5.5
(15)	Precipitation	PrecAvg	62.2	5.1	4.6	5.4	4.7	4.2	5.8	7.4	6.3	5.9	3.6	4.4
		PrecMax	86.5	15.8	9.8	11.6	17.0	13.3	13.9	15.7	10.0	22.8	12.6	11.7
		PrecMin	41.3	1.2	1.1	0.6	0.8	0.7	1.3	2.3	2.4	1.6	0.0	1.2
		PrecStd	11.6	3.2	2.1	3.0	3.4	2.8	3.2	2.9	2.4	4.2	3.1	3.0
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.8	76.9	79.3	81.8	90.9	97.2	95.7	95.4	92.5	89.5	80.3
			MCWB	68.6	70.1	68.2	69.1	74.4	76.9	78.5	80.3	77.0	76.7	71.1
		2%	DB	72.1	73.1	76.7	79.7	87.4	93.7	92.7	92.5	90.2	86.4	77.4
			MCWB	67.9	67.1	68.2	69.9	74.8	77.6	79.1	79.4	76.6	74.9	69.6
		5%	DB	70.0	70.5	73.7	78.3	84.9	90.9	90.8	90.7	88.6	84.0	75.2
			MCWB	66.6	66.1	67.2	69.5	73.9	76.3	78.4	79.0	76.6	74.1	68.8
		10%	DB	67.9	68.4	72.1	76.8	82.6	89.0	89.5	89.6	86.8	81.5	72.9
			MCWB	64.8	64.7	66.4	69.0	73.5	76.3	78.1	78.7	76.4	71.7	66.7
		0.4%	WB	70.9	72.1	72.6	76.8	82.7	83.3	84.4	80.7	79.3	75.2	74.0
			MCDB	71.9	74.5	75.6	78.3	83.7	90.7	90.6	90.5	87.2	84.4	77.1
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.0	69.9	70.9	74.6	77.5	81.2	81.9	83.0	79.5	77.7	72.7
			MCDB	70.5	71.7	73.8	77.2	82.5	87.9	88.9	89.4	85.8	83.4	75.6
		2%	WB	67.3	67.8	69.6	72.8	76.6	80.0	80.8	81.6	78.5	76.4	70.5
			MCDB	68.9	69.4	71.9	76.2	81.9	85.9	87.5	87.3	84.8	82.1	73.7
		5%	WB	65.6	65.9	68.3	71.4	75.4	78.8	79.9	80.6	77.9	74.8	68.4
			MCDB	67.8	67.6	70.7	74.5	81.0	84.7	86.3	86.2	84.2	80.1	71.8
		10%	WB	65.6	65.9	68.3	71.4	75.4	78.8	79.9	80.6	77.9	74.8	68.4
			MCDB	67.8	67.6	70.7	74.5	81.0	84.7	86.3	86.2	84.2	80.1	71.8
		0.4%	WB	70.9	72.1	72.6	76.8	82.7	83.3	84.4	80.7	79.3	75.2	74.0
			MCDB	71.9	74.5	75.6	78.3	83.7	90.7	90.6	90.5	87.2	84.4	77.1
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.3	18.5	18.0	17.3	16.5	15.3	14.6	14.2	15.4	19.5	21.0
			MCDBR	15.1	16.1	16.1	16.9	18.0	17.4	15.6	15.4	16.6	17.4	19.5
		5% WB	MCWBR	11.8	11.5	9.7	9.4	8.0	6.5	6.1	6.2	6.9	8.7	12.8
			MCWBR	13.6	13.6	13.4	12.8	13.9	14.0	14.3	13.9	14.1	14.7	17.0
			MCWBR	11.7	11.4	9.5	9.1	7.3	6.6	6.4	6.3	6.8	9.1	13.2
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.352	0.376	0.407	0.441	0.466	0.508	0.498	0.445	0.387	0.359	0.345
		taud	2.545	2.503	2.431	2.345	2.286	2.268	2.173	2.197	2.310	2.459	2.475	2.527
		Ebn at Noon	285	290	289	282	271	263	252	253	263	274	273	275
		Edh at Noon	27	31	36	40	43	43	48	46	40	32	29	27
(44)	All-Sky Solar Radiation	RadAvg	977	1167	1577	1905	2125	2006	1881	1753	1650	1449	1112	861
		RadStd	77	159	126	157	108	154	130	132	98	139	98	78

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	+288	+187

Nomenclature: See separate page